University Physics Quiz 6-2

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1. The tension force f in a rubber band when it is in equilibrium is given by

$$f = AT\left(\frac{l}{l_0} - \frac{l_0^2}{l^2}\right)$$

where A is a constant, T is the thermodynamic temperature, l is the length of the rubber band and l_0 is the length of the rubber band for $f = 0$. The change in volume of the rubber band is negligible. When $l = l_0$, the heat capacity of the rubber band at constant length is $C_l(T, l = l_0) = K = \text{Const.}$

(a)Show that: $(\partial U/\partial l)_T = 0$, $(\partial C_l/\partial l)_T = 0$.

(b) Find $C_l(T,l)$ and the internal energy $U(T,l)$ of the rubber band up to an additive constant.

(c) Find the entropy of the rubber band $S(T,l)$ up to an additive constant.

Solution: (a) The fundamental equation of the thermodynamics can be written as $dU = TdS + fdl$ or

$$dS = \frac{1}{T}dU - \frac{f}{T}dl\,.$$

Then, we have

$$\left(\frac{\partial S}{\partial T}\right)_l = \frac{1}{T}\left(\frac{\partial U}{\partial T}\right)_l, \quad \left(\frac{\partial S}{\partial l}\right)_T = \frac{1}{T}\left(\frac{\partial U}{\partial l}\right)_T - \frac{f}{T}$$

Making partial derivatives to l and T respectively to the two equations above, we obtain

$$\frac{\partial^2 S}{\partial l\partial T} = \frac{1}{T}\frac{\partial^2 U}{\partial l\partial T}, \quad \frac{\partial^2 S}{\partial T\partial l} = \frac{1}{T}\frac{\partial^2 U}{\partial T\partial l} - \frac{1}{T^2}\left(\frac{\partial U}{\partial l}\right)_T + \frac{f}{T^2} - \frac{1}{T}\left(\frac{\partial f}{\partial T}\right)_l$$

Using the integrability conditions $\frac{\partial^2 S}{\partial l\partial T} = \frac{\partial^2 S}{\partial T\partial l}$, $\frac{\partial^2 U}{\partial l\partial T} = \frac{\partial^2 U}{\partial T\partial l}$, we obtain

$$\left(\frac{\partial U}{\partial l}\right)_T = f - T\left(\frac{\partial f}{\partial T}\right)_l.$$

(Another way to reach the above equation is: From

$$dU = TdS + fdl = T\left(\frac{\partial S}{\partial T}\right)_l dT + \left(T\left(\frac{\partial S}{\partial l}\right)_T + f\right)dl$$

We have $\left(\frac{\partial U}{\partial l}\right)_T = T\left(\frac{\partial S}{\partial l}\right)_T + f$.

From $dF = d(U - TS) = -SdT + fdl$, we have $\left(\frac{\partial S}{\partial l}\right)_T = -\left(\frac{\partial f}{\partial T}\right)_l$ (a Maxwell relation).

We obtain $\left(\frac{\partial U}{\partial l}\right)_T = f - T\left(\frac{\partial f}{\partial T}\right)_l$.)

Substituting $f = AT\left(\frac{l}{l_0} - \frac{l_0^2}{l^2}\right)$ into above equation, we obtain $\left(\frac{\partial U}{\partial l}\right)_T = 0$. And obliviously, we have

$$\left(\frac{\partial C_l}{\partial l}\right)_T = \left[\frac{\partial}{\partial l}\left(\frac{\partial U}{\partial T}\right)_l\right]_T = \left[\frac{\partial}{\partial T}\left(\frac{\partial U}{\partial l}\right)_T\right]_l = 0.$$

(b) From $\left(\frac{\partial C_l}{\partial l}\right)_T = 0$, we know that C_l is the function of T only. Then

$$C_l(T,l) = C_l(T) = C_l(T, l = l_0) = K.$$

From $dU = \left(\frac{\partial U}{\partial T}\right)_l dT + \left(\frac{\partial U}{\partial l}\right)_T dl = C_l dT = KdT$

$$U(T,l) = U(T_0) + \int_{T_0}^{T} KdT = KT - KT_0 + U(T_0).$$

(c) From $dS = \frac{1}{T}dU - \frac{f}{T}dl = \frac{K}{T}dT - A\left(\frac{l}{l_0} - \frac{l_0^2}{l^2}\right)dl$, we obtain

$$S(T,l) = K\ln T - A\left(\frac{l^2}{2l_0} + \frac{l_0^2}{l}\right) + \text{Const.}$$

2. A two-dimensional ideal gas is a collection of N non-interacting point particles which are moving randomly in a planar or other two-dimensional space.

(a) Write down the Maxwell velocity distribution and the Maxwell speed distribution for the two-dimensional ideal gas.

(b) Find the most probable speed v_m, the average speed $\bar{v}$ and the root mean square speed v_{rms} of the particles in the two-dimensional ideal gas.

Solution: (a) For a classical two dimensional ideal gas the number of particles having the components of velocities between v_x and $v_x + dv_x$, v_y and $v_y + dv_y$ is

$$f(v_x, v_y)dv_x dv_y = N\left(\frac{m}{2\pi k_B T}\right)\exp\left(-\frac{1}{2}m(v_x^2 + v_y^2)/k_B T\right)dv_x dv_y,$$

where N is the total number of particles.

Then, the number of particles having speeds between v and $v + dv$ is

$$f(v)dv = N\left(\frac{m}{2\pi k_B T}\right)\exp\left(-\frac{1}{2}mv^2/k_B T\right)2\pi v dv.$$

(b) $\frac{df(v)}{dv} = 0 \Rightarrow \quad \frac{d}{dv}\left(ve^{-\frac{mv^2}{2k_BT}}\right) = e^{-\frac{mv^2}{2k_BT}} + v\left(-\frac{mv}{k_BT}\right)e^{-\frac{mv^2}{2k_BT}} = 0 \Rightarrow \quad 1 - \frac{mv^2}{k_BT} = 0 \Rightarrow v_m = \sqrt{\frac{k_BT}{m}}.$

$$\bar{v} = \frac{\int_0^\infty vf(v)dv}{N} = \frac{m}{2\pi k_B T}\int_0^\infty v\exp\left(-\frac{1}{2}mv^2/k_BT\right)2\pi v dv = \frac{m}{k_BT}\int_0^\infty \exp\left(-\frac{1}{2}mv^2/k_BT\right)v^2 dv$$

$$= \frac{m}{k_BT}\left(\frac{2k_BT}{m}\right)^{\frac{3}{2}}\int_0^\infty x^2 e^{-x^2}dx = \sqrt{\frac{8k_BT}{m}}I_2 = \sqrt{\frac{8k_BT}{m}}\frac{\sqrt{\pi}}{4} = \sqrt{\frac{\pi k_BT}{2m}}.$$

$$\overline{v^2} = \frac{\int_0^\infty v^2 f(v)dv}{N} = \frac{m}{2\pi k_BT}\int_0^\infty v^2\exp\left(-\frac{1}{2}mv^2/k_BT\right)2\pi v dv$$

$$= \frac{m}{k_BT}\int_0^\infty \exp\left(-\frac{1}{2}mv^2/k_BT\right)v^3 dv = \frac{m}{k_BT}\left(\frac{2k_BT}{m}\right)^2\int_0^\infty x^3 e^{-x^2}dx = \frac{4k_BT}{m}I_3 = \frac{4k_BT}{m}\cdot\frac{1}{2} = \frac{2k_BT}{m}$$

$$v_{rms} = \sqrt{\frac{2k_BT}{m}}.$$

$\left(I_{2m} = \int_0^\infty x^{2m}e^{-x^2}dx = \frac{1}{2}\frac{1}{2}\frac{3}{2}\cdots\frac{2m-1}{2}\sqrt{\pi}, \quad I_{2m-1} = \frac{1}{2}(m-1)!.\right)$

3. (a) An ideal gas is kept in a container of volume V at pressure p and temperature T. Its molecules of mass m obey the Maxwell distribution. Suppose there is a small hole in the wall of the container of area A, and the gas is

escaping into the vacuum surrounding the container.
(a) Find the rate of the escape.

(b)Show that time-dependence of the pressure in the container is given by $p(t) = p_0 e^{-t/\tau}$ where $\tau = \frac{V}{A}\sqrt{2\pi m/k_B T}$ and p_0 is the pressure of the gas at $t = 0$.

Solution: Choose z-direction to be perpendicular to the wall with the hole. If a molecule has velocity v and its z-component is v_z (v_z), it will hit the wall during the time dt if it is no farther than $v_z dt$ from the wall. Thus , the total number of molecules with velocities within $(v_x, v_y, v_z) \to (v_x + dv_x, v_y + dv_y, v_z + dv_z)$ that will hit the area A during the time dt is

$$A v_z dt \frac{N}{V}\left(\frac{m}{2\pi k_B T}\right)^{3/2} \mathrm{e}^{-\frac{mv^2}{2k_B T}} dv_x dv_y dv_z.$$

Therefore, the total number of molecules with that will hit the area A on the wall during the time dt is

$$\begin{aligned} dN &= \int_{-\infty}^{\infty} dv_x \int_{-\infty}^{\infty} dv_y \int_0^{\infty} dv_z A v_z dt \frac{N}{V}\left(\frac{m}{2\pi k_B T}\right)^{3/2} \mathrm{e}^{-\frac{m(v_x^2+v_y^2+v_z^2)}{2k_B T}} \\ &= \frac{AN}{V} dt \left(\frac{m}{2\pi k_B T}\right)^{1/2} \int_0^{\infty} dv_z v_z \mathrm{e}^{-\frac{mv_z^2}{2k_B T}} \\ &= \frac{AN}{V} dt \left(\frac{m}{2\pi k_B T}\right)^{1/2} \frac{k_B T}{m} \int_0^{\infty} d\left(\frac{mv_z^2}{2k_B T}\right) \mathrm{e}^{-\frac{mv_z^2}{2k_B T}} \end{aligned}$$

Therefore

$$\frac{dN}{dt} = \frac{AN}{V}\left(\frac{k_B T}{2\pi m}\right)^{\frac{1}{2}} = AN \frac{p}{N k_B T}\left(\frac{k_B T}{2\pi m}\right)^{\frac{1}{2}} = \frac{Ap}{(2\pi m k_B T)^{\frac{1}{2}}}.$$

(b) From the equation of state $p = \frac{N k_B T}{V}$, we have $\frac{dp}{dt} = \frac{k_B T}{V}\frac{dN}{dt}$. Using the result of (a), we have

$$\frac{dp}{dt} = -\frac{A}{V}\left(\frac{k_B T}{2\pi m}\right)^{\frac{1}{2}} p = -\frac{p}{\tau}$$

where $\tau = \frac{V}{A}\sqrt{2\pi m/k_B T}$. Rewriting the above equation as

$$\frac{dp}{p} = -\frac{dt}{\tau}.$$

and integrating both side of above equation, we have

$$\int_{p_0}^{p} \frac{dp}{p} = -\int_0^t \frac{dt}{\tau} \quad \text{or} \quad \ln\frac{p}{p_0} = -\frac{t}{\tau} \quad \text{or} \quad p(t) = p_0 e^{-t/\tau}.$$

4. An ideal gas is composed of two type of molecules A and B whose molecular masses are m_1 and m_2 respectively. The number densities of two kinds of molecules are n_1 and it n_2 respectively. The gas is in equilibrium state of temperature T. Show that the average relative speed of a A molecule to a B molecule is

$$\langle v_{BA} \rangle = \sqrt{\frac{8k_B T}{\pi\mu}}$$

where $\mu = m_1 m_2/(m_1 + m_2)$ is the reduced mass **.**

Solution: The probability of an A molecule in the volume element $dv_{Ax}dv_{Ay}dv_{Az}$ in the vicinity of $\boldsymbol{v}_A$ in the velocity space and a B molecule in the volume element $dv_{Bx}dv_{By}dv_{Bz}$ in the vicinity of $\boldsymbol{v}_B$ in the velocity space can be written as

$$P(v_{Ax},v_{Ay},v_{Az})P(v_{Bx},v_{By},v_{Bz})dv_{Ax}dv_{Ay}dv_{Az}dv_{Bx}dv_{By}dv_{Bz} =$$

$$= C_1 \exp\left[-\frac{m_1\left(v_{Ax}^2+v_{Ay}^2+v_{Az}^2\right)}{2k_BT}\right] C_2 \exp\left[-\frac{m_2\left(v_{Bx}^2+v_{By}^2+v_{Bz}^2\right)}{2k_BT}\right] dv_{Ax}dv_{Ay}dv_{Az}dv_{Bx}dv_{By}dv_{Bz}$$

where $C_1 = \left(\frac{m_1}{2\pi k_BT}\right)^{\frac{3}{2}}$ and $C_2 = \left(\frac{m_2}{2\pi k_BT}\right)^{\frac{3}{2}}$ are normalizing factors so that

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} C_1 \exp\left[-\frac{m_1\left(v_{Ax}^2+v_{Ay}^2+v_{Az}^2\right)}{2k_BT}\right] dv_{Ax}dv_{Ay}dv_{Az} = 1$$

and

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} C_2 \exp\left[-\frac{m_2\left(v_{Bx}^2+v_{By}^2+v_{Bz}^2\right)}{2k_BT}\right] dv_{Bx}dv_{By}dv_{Bz} = 1.$$

Introducing the relative velocity $\boldsymbol{v}_r = \boldsymbol{v}_A - \boldsymbol{v}_B$ and the center-of-mass velocity $\boldsymbol{V}_C = \frac{m_1\boldsymbol{v}_A + m_2\boldsymbol{v}_B}{m_1+m_2}$, we have

$$dV_{Cx}dV_{Cy}dV_{Cz}dv_{rx}dv_{ry}dv_{rz} = J dv_{Ax}dv_{Ay}dv_{Az}dv_{Bx}dv_{By}$$

where J is the Jocabian determinant

$$J = \det\begin{vmatrix}
\frac{\partial V_{Cx}}{\partial v_{Ax}} & \frac{\partial V_{Cx}}{\partial v_{Ay}} & \frac{\partial V_{Cx}}{\partial v_{Az}} & \frac{\partial V_{Cx}}{\partial v_{Bx}} & \frac{\partial V_{Cx}}{\partial v_{By}} & \frac{\partial V_{Cx}}{\partial v_{Bz}} \\
\frac{\partial V_{Cy}}{\partial v_{Ax}} & \frac{\partial V_{Cy}}{\partial v_{Ay}} & \frac{\partial V_{Cy}}{\partial v_{Az}} & \frac{\partial V_{Cy}}{\partial v_{Bx}} & \frac{\partial V_{Cy}}{\partial v_{By}} & \frac{\partial V_{Cy}}{\partial v_{Bz}} \\
\frac{\partial V_{Cz}}{\partial v_{Ax}} & \frac{\partial V_{Cz}}{\partial v_{Ay}} & \frac{\partial V_{Cz}}{\partial v_{Az}} & \frac{\partial V_{Cz}}{\partial v_{Bx}} & \frac{\partial V_{Cz}}{\partial v_{By}} & \frac{\partial V_{Cx}}{\partial v_{Bz}} \\
\frac{\partial v_{rx}}{\partial v_{Ax}} & \frac{\partial v_{rx}}{\partial v_{Ay}} & \frac{\partial v_{rx}}{\partial v_{Az}} & \frac{\partial v_{rx}}{\partial v_{Bx}} & \frac{\partial v_{rx}}{\partial v_{By}} & \frac{\partial v_{rx}}{\partial v_{Bz}} \\
\frac{\partial v_{ry}}{\partial v_{Ax}} & \frac{\partial v_{ry}}{\partial v_{Ay}} & \frac{\partial v_{ry}}{\partial v_{Az}} & \frac{\partial v_{ry}}{\partial v_{Bx}} & \frac{\partial v_{ry}}{\partial v_{By}} & \frac{\partial v_{ry}}{\partial v_{Bz}} \\
\frac{\partial v_{rz}}{\partial v_{Ax}} & \frac{\partial v_{rz}}{\partial v_{Ay}} & \frac{\partial v_{rz}}{\partial v_{Az}} & \frac{\partial v_{rz}}{\partial v_{Bx}} & \frac{\partial v_{rz}}{\partial v_{By}} & \frac{\partial v_{rz}}{\partial v_{Bz}}
\end{vmatrix}$$

$$= \begin{vmatrix}
\frac{m_1}{m_1+m_2} & 0 & 0 & \frac{m_2}{m_1+m_2} & 0 & 0 \\
0 & \frac{m_1}{m_1+m_2} & 0 & 0 & \frac{m_2}{m_1+m_2} & 0 \\
0 & 0 & \frac{m_1}{m_1+m_2} & 0 & 0 & \frac{m_2}{m_1+m_2} \\
1 & 0 & 0 & -1 & 0 & 0 \\
0 & 1 & 0 & 0 & -1 & 0 \\
0 & 0 & 1 & 0 & 0 & -1
\end{vmatrix} = 1.$$

Noting that

$$\frac{1}{2}m_1v_A^2 + \frac{1}{2}m_2v_B^2 = \frac{1}{2}(m_1+m_2)V_C^2 + \frac{1}{2}\mu v_r^2$$

where $\mu = m_1m_2/(m_1+m_2)$ is the reduced mass, we can write

$$P(v_{Ax},v_{Ay},v_{Az})P(v_{Bx},v_{By},v_{Bz})dv_{Ax}dv_{Ay}dv_{Az}dv_{Bx}dv_{By}dv_{Bz} =$$

$$= \left(\frac{m_1}{2\pi k_BT}\right)^{\frac{3}{2}}\left(\frac{m_2}{2\pi k_BT}\right)^{\frac{3}{2}} \exp\left[-\frac{(m_1+m_2)V_C^2}{2k_BT}\right]\exp\left(-\frac{\mu v_r^2}{2k_BT}\right) dV_{Cx}dV_{Cy}dV_{Cz}dv_{rx}dv_{ry}dv_{rz}.$$

Integrating the above expression over the center-of mass velocity space and noting that

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\exp\left[-\frac{(m_1+m_2)V_C^2}{2k_BT}\right]dV_{Cx}dV_{Cy}dV_{Cz} = \left(\frac{2\pi k_BT}{m_1+m_2}\right)^{\frac{3}{2}},$$

we obtain

$$P(v_{rx},v_{ry},v_{rz})dv_{rx}dv_{ry}dv_{rz}$$

$$=\left(\frac{\frac{m_1m_2}{m_1+m_2}}{2\pi k_BT}\right)^{\frac{3}{2}}\exp\left(-\frac{\mu v_r^2}{2k_BT}\right)dv_{rx}dv_{ry}dv_{rz} = \left(\frac{\mu}{2\pi k_BT}\right)^{\frac{3}{2}}\exp\left(-\frac{\mu v_r^2}{2k_BT}\right)dv_{rx}dv_{ry}dv_{rz}.$$

Using the spherical coordinates for the relative velocity with its volume element as

$$v_r^2\sin\theta dv_r d\theta d\phi$$

and integrating over the angular coordinates ($\int_0^{\pi}\int_0^{2\pi}\sin\theta d\theta d\phi = 4\pi$), we have the probability distribution of relative speed

$$P(v_r)dv_r$$

$$=\left(\frac{\mu}{2\pi k_BT}\right)^{\frac{3}{2}}4\pi v_r^2\exp\left(-\frac{\mu v_r^2}{2k_BT}\right)dv_r.$$

Therefore, the average relative speed is

$$\langle v_{BA}\rangle = \int_0^{\infty}v_rP(v_r)dv_r$$

$$=\left(\frac{\mu}{2\pi k_BT}\right)^{\frac{3}{2}}4\pi\int_0^{\infty}v_r^3\exp\left(-\frac{\mu v_r^2}{2k_BT}\right)dv_r = \sqrt{\frac{8k_BT}{\pi\mu}}.$$